

Spanning-tree extrema in $3 \times n$ triangulated grids for $n = 3, 4, 5$ and a minimal five-state transfer

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Abstract

Place one of the two diagonals in every cell of a $3 \times n$ rectangular grid. We determine the minimum and maximum numbers of ordinary spanning trees, and every equality configuration up to geometric symmetry, for $n = 3, 4, 5$. The respective minimum–maximum pairs are (1792, 2205), (65431, 82368), and (2248641, 3057118). Each extremum has one geometric orbit. A boundary-forest argument gives four explicit 5×5 integer matrices that count spanning trees for every prescribed sequence of diagonal choices. A nonzero 5×5 count determinant proves that five is the minimum dimension among uniform rational linear matrix representations of this word function. The finite classification exhausts all 336 configurations using exact arithmetic. The extremal problem for general n remains unresolved here.

1 The finite classification

For $n \geq 1$, let the vertices be (r, c) with $1 \leq r \leq 3$ and $1 \leq c \leq n$. Include every nearest-neighbor horizontal and vertical edge. Each cell (r, c) , where $r \in \{1, 2\}$ and $1 \leq c < n$, receives exactly one diagonal: bit 0 selects $\{(r, c), (r+1, c+1)\}$, and bit 1 selects $\{(r+1, c), (r, c+1)\}$. Read the cell bits row by row, top row first and left to right, to form $\sigma \in \{0, 1\}^{2(n-1)}$. Write $G_n(\sigma)$ for the resulting simple graph and $\tau(G)$ for its number of ordinary, unweighted spanning trees.

Dhokal, Nguyen and Wu study triangular strips of width two and ask about the effect of diagonal choices at larger widths [1, Section 5, pp. 21–22]. Their two illustrated 3×3 configurations have 2146 and 2080 spanning trees; they are examples, not asserted extrema. With rows increasing downward, their Figures 22 and 23 have encodings 0100 and 0000, respectively. Our counts reproduce both. The present question concerns ordinary spanning trees, irrespective of whether a tree is balanced.

For $n = 3$ the permitted equivalences are the eight geometric symmetries of the square. For $n = 4, 5$ they are the four geometric symmetries of the rectangle. In both cases the action is induced by vertex coordinates and therefore by the images of diagonal edges. Orbit sizes count distinct configurations; equivalence under arbitrary abstract graph isomorphisms is not used.

Theorem 1. *For $n \in \{3, 4, 5\}$, the complete extremal classification is given in Table 1. Each row of the table describes exactly one geometric orbit. The full attaining sets are*

$$\begin{aligned} E_3^- &= \{1001\}, & E_3^+ &= \{0110\}, \\ E_4^- &= \{100011, 110001\}, & E_4^+ &= \{001100, 011110\}, \\ E_5^- &= \{10100101\}, \end{aligned}$$

$$E_5^+ = \{00011100, 00111000, 00111110, 01111100\}.$$

Here E_n^- and E_n^+ denote the configurations attaining the minimum and maximum, respectively.

Table 1: All extremal geometric orbits. Representatives are lexicographically least within their orbits.

n	Extremum	Tree count	Representative	Orbit size
3	Minimum	1792	1001	1
3	Maximum	2205	0110	1
4	Minimum	65431	100011	2
4	Maximum	82368	001100	2
5	Minimum	2248641	10100101	1
5	Maximum	3057118	00011100	4

The counting mechanism below specializes the established boundary-partition method of Raff [2, Section 5] to two independently oriented diagonals per slab. The contribution relative to those inspected sources is the complete finite equality classification, the four explicit orientation-dependent matrices, and the dimension certificate of Theorem 3. The general transfer principle and the matrix-rank argument are established methods. For broader adjacent methodology involving exact finite certification in enumerative combinatorics, see Zeng [3]; that reference is not a premise for any of the spanning-tree results here. These comparisons do not assert priority or an exhaustive literature determination.

2 A forest transfer for every orientation word

Retain a spanning forest of a j -column prefix when every component meets column j . Its connectivity on the three boundary vertices is one of the partitions, in the following fixed order:

$$\mathcal{P} = (123, 12|3, 13|2, 1|23, 1|2|3). \quad (1)$$

A block is exactly the boundary intersection of one forest component. Let f_j be the column vector of retained-forest counts in this order. Isolated boundary vertices are allowed. The four subsets of the two edges in the initial three-vertex path give

$$f_1 = (1, 1, 0, 1, 1)^T. \quad (2)$$

Although state $13|2$ is initially absent, later slabs can produce it. State 123 consists exactly of spanning trees, since every component must meet the boundary.

Label the old boundary vertices $0, 1, 2$ and the new ones $3, 4, 5$, each from top to bottom. If the two cells in a new slab have top and bottom bits a, b , the seven new edges, in this order, are

$$(3, 4), (4, 5), (0, 3), (1, 4), (2, 5), d_a, e_b, \quad (3)$$

where

$$d_0 = (0, 4), \quad d_1 = (1, 3), \quad e_0 = (1, 5), \quad e_1 = (2, 4).$$

Given an input partition p , identify old boundary vertices in the same block and leave the new boundary vertices distinct. For each of the 128 subsets S of (3), reject S if the quotient multigraph has a cycle or a component missing the new boundary. Otherwise let q be its partition on $3, 4, 5$. Define $M_{ab}[q, p]$ as the number of accepted subsets yielding q . Parallel edges remain distinct in the cycle test.

Proposition 2. *For every prescribed sequence of slabs, the retained-forest vectors satisfy $f_{j+1} = M_{ab}f_j$ at a slab with bits a, b . In the state order (1), with output states indexing rows, the four matrices are*

$$M_{00} = \begin{pmatrix} 19 & 12 & 21 & 15 & 9 \\ 7 & 6 & 8 & 9 & 6 \\ 2 & 1 & 2 & 0 & 0 \\ 9 & 6 & 11 & 10 & 6 \\ 5 & 4 & 6 & 6 & 4 \end{pmatrix}, \quad M_{01} = \begin{pmatrix} 16 & 15 & 12 & 15 & 9 \\ 8 & 10 & 6 & 9 & 6 \\ 1 & 0 & 1 & 0 & 0 \\ 8 & 9 & 6 & 10 & 6 \\ 5 & 6 & 4 & 6 & 4 \end{pmatrix}, \quad (4)$$

$$M_{10} = \begin{pmatrix} 21 & 13 & 24 & 13 & 8 \\ 7 & 6 & 8 & 5 & 4 \\ 4 & 2 & 5 & 2 & 1 \\ 7 & 5 & 8 & 6 & 4 \\ 5 & 4 & 6 & 4 & 3 \end{pmatrix}, \quad M_{11} = \begin{pmatrix} 19 & 15 & 21 & 12 & 9 \\ 9 & 10 & 11 & 6 & 6 \\ 2 & 0 & 2 & 1 & 0 \\ 7 & 9 & 8 & 6 & 6 \\ 5 & 6 & 6 & 4 & 4 \end{pmatrix}. \quad (5)$$

Proof. Fix a retained old forest. Contract each of its tree components. Only old boundary vertices can have incident new edges, so the contracted extension depends on the old forest solely through p . Contracting tree edges preserves cycle rank. In particular, the union of the old forest and S is acyclic exactly when the quotient multigraph is a forest. One may test this by $|S| = |V| - c$, with V the quotient vertex set and c its number of components, including isolated vertices. Edge multiplicities must be counted. The second rejection test is precisely the retained-forest condition at the new boundary.

Conversely, every retained forest on $j + 1$ columns restricts to a retained forest on j columns. If a component of the restriction missed column j , no new edge could touch it. It would remain a component missing column $j + 1$, a contradiction. The restriction and its subset of new edges are recovered uniquely from the full forest. Thus old retained forests paired with accepted subsets are in bijection with new retained forests, proving the recurrence.

Enumerating the $4 \cdot 5 \cdot 128 = 2560$ orientation–partition–subset cases in the definition gives (4)–(5). The enumeration uses only the seven listed edges and the five partitions; the accompanying verifier reconstructs every entry by the component edge-count test. As a small check, $M_{01}[13|2, 123] = 1$: the unique accepted subset is $\{(0, 3), (2, 5)\}$, mask 20 when edge k in (3) has weight 2^{k-1} . The new top and bottom vertices are connected through the old component, and the middle vertex is isolated. $\square$

Let $\mathcal{A} = \{00, 01, 10, 11\}$ be the slab alphabet. For a word $w = a_1 \cdots a_m \in \mathcal{A}^*$, let $F(w)$ be the tree count of its $(m + 1)$ -column graph. The empty word ε represents the initial path, so $F(\varepsilon) = 1$. With $e_1 = (1, 0, 0, 0, 0)^\top$, Proposition 2 proves, for every finite word,

$$F(a_1 \cdots a_m) = e_1^\top M_{a_m} \cdots M_{a_1} f_1. \quad (6)$$

This universal formula follows from the forest bijection, independently of the finite range used for optimization.

3 Exhaustion and geometric equality cases

Proof of Theorem 1. Put $m = n - 1$. A row-major string σ gives slab letters $a_{c+1} = \sigma_c \sigma_{m+c}$ for $0 \leq c < m$, using zero-based string indices. All $2m$ bits are chosen independently, so enumerating every binary string visits every configuration exactly once. No symmetry reduction is used in the count computation.

Apply (6) to each string using integer arithmetic. The following complete enumeration procedure uses the displayed matrices; `product` is the Cartesian-product function in Python's standard `itertools` module.

```

for n in (3, 4, 5):
    m = n - 1
    counts = {}
    for b in product("01", repeat=2*m):
        v = [1, 1, 0, 1, 1]
        for c in range(m):
            A = M[b[c] + b[m+c]]
            v = [sum(x*y for x, y in zip(row, v)) for row in A]
        counts["".join(b)] = v[0]
    assert len(counts) == 4**m
    low, high = min(counts.values()), max(counts.values())
    Eminus = sorted(b for b in counts if counts[b] == low)
    Eplus = sorted(b for b in counts if counts[b] == high)

```

The respective domain sizes are 16, 64 and 256. The values `low`, `high`, `Eminus`, and `Eplus` are exactly those stated in the theorem. The accompanying verifier asserts equality with all six extrema and all six full attaining sets, and outputs the entire table of 336 counts.

It remains to identify the geometric orbits within these complete sets. View a configuration as a $2 \times m$ bit matrix B with indices $r \in \{0, 1\}$ and $0 \leq c < m$. Reflection of vertex rows and reflection of vertex columns act, respectively, by

$$R(B)_{r,c} = 1 - B_{1-r,c}, \quad C(B)_{r,c} = 1 - B_{r,m-1-c}. \quad (7)$$

The complements reflect reversal of diagonal slope. At $n = 3$, coordinate transposition gives the further action

$$T(B)_{r,c} = B_{c,r}. \quad (8)$$

These formulas follow directly by mapping the two endpoints of a diagonal. The first two generate the rectangle group; adding T for the square generates its eight geometric symmetries. Taking closure of each representative in Table 1 under these generators gives exactly the corresponding set E_n^- or E_n^+ , with cardinalities 1, 1, 2, 2, 1, 4. Each such set is therefore one full orbit, and the exhaustion already excludes any additional equality configurations. $\square$

As a separate corroboration, all 336 counts agree with reduced-Laplacian determinants, evaluated both by integer Bareiss elimination and by rational Gaussian elimination in separately implemented programs. A direct enumeration of all $\binom{16}{8} = 12870$ candidate tree-edge subsets for each of the 16 three-column graphs also agrees, totaling 205920 connectivity tests. These checks complement the forest proof and the complete finite enumeration; they are not extrapolations to larger n .

4 The minimum linear dimension is five

A *uniform rational linear representation of dimension d* for F consists of fixed vectors $u, \lambda \in \mathbb{Q}^d$ and one fixed matrix $A_a \in \mathbb{Q}^{d \times d}$ per letter $a \in \mathcal{A}$, such that

$$F(a_1 \cdots a_m) = \lambda^\top A_{a_m} \cdots A_{a_1} u \quad \text{for every } m \geq 0. \quad (9)$$

The vectors and matrices are independent of both word length and letter position. The case $m = 0$ uses the identity matrix.

Theorem 3. *The minimum dimension in (9) is five. Agreement with F on all words of length at most two already forces dimension at least five.*

Proof. Equation (6) supplies a representation of dimension five. For the lower bound, use prefixes p and suffixes s from the ordered list $(\varepsilon, 00, 01, 10, 11)$ and form $H_{s,p} = F(ps)$. Thus rows index suffixes and columns index prefixes. The counts give

$$H = \begin{pmatrix} 1 & 55 & 55 & 55 & 55 \\ 55 & 2080 & 2146 & 1972 & 2071 \\ 55 & 1972 & 2071 & 1792 & 1972 \\ 55 & 2146 & 2205 & 2071 & 2146 \\ 55 & 2071 & 2146 & 1972 & 2080 \end{pmatrix}, \quad \det H = -421362 \neq 0. \quad (10)$$

Every entry concerns a word of length at most two. The determinant is an exact integer calculation; it is reproduced by the verifier.

For any representation (9), abbreviate the reversed matrix product for a word p as A_p , with $A_\varepsilon = I$. Let the p -th column of U be $A_p u$, and the s -th row of L be $\lambda^\top A_s$. Concatenation gives $H_{s,p} = \lambda^\top A_s A_p u$, so $H = LU$. Consequently $\text{rank } H \leq d$. Since (10) has rank five, $d \geq 5$, establishing both assertions. $\square$

This lower bound applies precisely to the fixed-vector, fixed-per-letter rational linear model. It does not address nonlinear encodings, length-dependent or position-dependent matrices, affine models with another dimension convention, or restricted classes of orientation words. It yields no general-length extremal formula.

5 Reproducibility and scope

The accompanying file `verify.py` uses only the Python standard library. Running

```
python3 manuscript/verify.py > manuscript/verification.json
```

from the source-package root reconstructs all four matrices by enumerating the local edge subsets, enumerates the 336 configurations, checks the geometric orbits and the two source examples, and evaluates (10). It prints the complete count tables and a summary of the checks. The script contains the displayed matrices and expected equality sets explicitly, so it checks these transcriptions as well as recomputing the outputs. All arithmetic is exact. The retained research certificates additionally record every accepted local subset and the separately implemented determinant and direct-tree checks.

The extrema proved here cover exactly $n = 3, 4, 5$. The recurrence counts any prescribed orientation word, but optimization over words of arbitrary length is not settled. The comparison with prior work is limited to the inspected primary sections of Dhakal–Nguyen–Wu and Raff and a retained arXiv search record inspected on 9 September 2026. No exhaustive nonduplication or priority claim is made.

Computational and AI assistance. AI assistance was used to develop the proof exposition, implement the exact enumerations and certificate checks, and prepare this manuscript. A separate AI review context reconstructed the local transfers, checked all 336 counts and the geometric equality sets, and recomputed the dimension certificate with separately written code. That review is not external human peer review or statistical independence. The mathematical claims rest on the stated arguments and reproducible exact calculations.

References

- [1] S. Dhakal, L. Nguyen and Y. Wu, *Balanced spanning trees for triangular strip lattices*, arXiv:2609.08222v1 (2026). <https://arxiv.org/abs/2609.08222v1>.
- [2] P. Raff, *Spanning Trees in Grid Graphs*, arXiv:0809.2551v1 (2008). <https://arxiv.org/abs/0809.2551v1>.
- [3] Z. Zeng, *The Distinct-Cycle-Length Probability is Strictly Decreasing after 30*, SSRN working paper (2026). <https://doi.org/10.2139/ssrn.7380519>.